

Mobile watch

Google's Android, not the iPhone OS is predicted to be one of the top contenders in the mobile OS war

22

Web watch

With Google Wave invites being given away, is it really as useful as everyone wishes it to be?

26



What will Web 3.0 have in store for us? The most educated guesses will be short-sighted, but there are a few tantalizing hints from the bleeding edge of Web 2.0 development.

Aditya Madanapalle

aditya.madanapalle@thinkdigit.com

Think of the web as an ecosystem, a constantly evolving, organic and volatile entity. Just as real world ecosystems have eras, the web too goes through phases. The first phase was the e-era or the dot com era. It became commercially viable to offer content and services online, and a big bunch of companies that were known primarily by their internet addresses came up. This stage of the web was also known as the “bubble”. Web 2.0, came into existence roughly in the second decade of the internet, and this was the “cloud” stage. User interaction, social networking, and user generated content are some of the hallmarks of Web 2.0. Right now, we are on the verge of another era, moving beyond what Web 2.0 has to offer. Tentatively called Web 3.0, exciting times lie ahead, where the web will get more firmly ingrained as a part of our lives. Most of the technology used in Web 2.0, the wikis, the widgets, and the APIs, were around on little used sites, and on design boards long before the onset of Web 2.0. In fact, some of these ideas were around since before the internet itself came into being, let alone the web. To get a clear idea of where the web is headed, it is necessary to have a deeper understanding of the technology that drives it. On the way to Web 3.0, users have conquered massive digital divides, survived a massive bubble burst, and even endured a full fledged war.

Laying the foundations

The web is made up of many layers of standards and protocols. Assume ten computers are connected to each other, and needed to communicate with each other. The simplest solution was in the wiring itself. Ask for something in a remote location, and you got it as if you were navigating to another location in your

own memory. If you have ten clusters of ten computers each, depending on the wires was not feasible. The answer to this was to make each computer a host, identify it with a number, and allow for navigation using these numbers. This was a protocol that laid the foundations to the internet, and was called TCP/IP. These protocols themselves have versions numbers, and currently version 4 is being used. The next generation will be version 6 that will allow for more computers to be interconnected. The number of digits available in the original standards have been exceeded by the growth of the internet. Now you have hundreds of computers, connected to each other, each with a uniquely identifiable number on the network. What then?

Each computer, or host, on the network, had something to offer, usually in the form of data. One solution was to give access to documents in whatever format they were created in. This would, however, cause a lot of problems. For one, not everyone could read all the formats of all the documents. This would mean that every user would have to install a large number of document readers, which was not practical at all. Moreover, this would heavily eat up the limited bandwidth available. The other solution was to represent all the data in a common and simple language, as close to plain text as possible, and have a client software interpret the data to display on screen. This was the approach taken for the web, the client software was a web browser, and the data was represented then in HTML. Hyper Text Markup Language was an innovative answer. Tags owe their existence to these early days of the web. Each line in a plain text document was tagged in a way that allowed a browser to interpret the tag, and display the text attached to it, on screen. This was a simple, and robust enough system. Anyone could set up a host, write in a few lines of code, and offer their content and information online for the world to see. There were however, a few challenges at this juncture of the internet.

**Bridging the great divide**

The first challenge was to get people online at all. The network of computers had to spread slowly from continent to continent and country to country. Phone lines were initially used for this, soon to be replaced by a massive grid of fibre-optic lines. The next challenge was to standardise the HTML language. This led to the browser wars, where Netscape, and IE would render the tags slightly differently, and sometimes support tags that other browsers did not. So if someone who created a web site wanted text to scroll across the screen in a marquee, one browser allowed users to view it as such, but the other did not. The browser wars have toned down a bit now, but are still an ongoing source of irritation for the users of the web. Another important breakthrough necessary was something to make sense of, and consolidate all the information on the web. Imagine going online, without Google, if you can. That is how the web was for quite some time. Without a search engine to direct you to the information that you wanted, the web was a meaningless heap of content that could not be navigated. The only way to get anywhere before search engines was through portals, that heavily selected the direction of your surfing. However, crawlers that jumped from blue lines to blue lines, and indexed all the information on the web came into being, allowing for a window into the wide and complex web.

The web back then was basically for information. You wanted to know about something, you went online, searched for it, browsed through it, and logged off. The users were spectators, without any real active participation. People who generated content did so on their own home pages. Email forwards had their day, one of the few ways that people could actually reach out to other people and say “hey! check this out! this is interesting”. We could have actually used 1GB mailboxes back then. Success stories such as Usenet and Napster were few and far between,